Development of an Image Processing System for Plant Phenotyping in Low-cost Automated Greenhouse Environments

by

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Abstract

Human population growth continues to strain the global food supply, and agricultural production needs to increase by 70% by 2050 to meet this demand. As a result, the development of cultivars that can produce the most yield and are resistant to various adverse abiotic and biotic stressors has become increasingly important. Plant phenotyping methods are employed to investigate gene-environment interactions in the development of such cultivars, and over the past decade, automated image-processing and analysis techniques in plant phenotyping have assisted in these investigations. Despite that, automated image-based tasks can become costly. Consequently, there is a need for the development of low-cost, automated image-based plant phenotyping systems in greenhouse environments. This study aims to develop an open-source image-processing-based plant phenotyping component, which helps to detect various factors of plant health, stress and growth in order to support low-cost automated greenhouses.

Contents

1 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [1], which is important in solving the problem of providing an adequate global food supply. The rapid growth of the human population has created a strain on international food security. It is estimated that crop production must increase by 70% by 2050 [2] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [3]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximize yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been an effective tool for plant breeding [4].

1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [5], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [6]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [7, 8, 9].

1.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [5] for the development of germplasm with improved plant traits [9]. The data produced by phenotyping is used as a proxy for assessing a plant's performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have a complex dynamic feedback loop with their environment [10]. Therefore phenotypical traits serve as proxies in determining whether certain genotypes have favourable traits in particular environmental conditions.

1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [11]. Holistic properties consider the plant as a whole, whereas component-based features look at organs like the organism's fruit, flower, leaves, stem, etc.

Leaf-derived traits are one of the most commonly used plant status indicators because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth and stress [12].Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [13]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [14, 12], and leaf thickness for measuring abiotic stressors like water scarcity [15]. Additionally, physiological traits of leaves exist that indicate various factors of plant status like stomatal conductance for the detection of biotic stressors like pathogens [16], leaf water content for drought stress [17].

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [12]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [18], stems [19], seeds [20], and fruit [21].

2 Image-Based Plant Phenotyping

Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant's ontogenesis [22, 12, 23]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which are amenable to automation and can analyse large samples of data precisely in a very short period. Reducing the need for potentially erroneous, time-consuming, expensive human labour in plant phenotyping [11].

These Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems [21], which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [24]. For image-based plant phenotyping, this means that image acquisition and processing techniques are vital.

2.1 Acquistion Techniques

Image processing techniques in plant phenotyping cannot be applied until suitable images are acquired. Image sensor advancements have provided the foundations for detecting various information from plants using electromagnetic radiation. Plant tissues interact with this radiation by absorbing, reflecting, emitting, transmitting, and fluorescing light. Different ranges of wavelengths correspond to different biochemical or physiological aspects of plant tissue, which are not visible to the naked eye [9]. Thus, the choice of imaging technique for data acquisition affects the types of phenotypical parameters that can be measured.

Current imaging techniques are visible, fluorescence, thermal, tomographic, and hyperspectral imaging [25, 9], where the most commonly used techniques are visible, and hyperspectral imaging due to their affordablity, robustness and flexibility in measuring a wide range of phenotypic parameters. The most common imaging technique is visible-imaging. Typically afforadable, and high-resolution RGB cameras are used in visible imaging. These devices are sensitive to the visible spectral range and have been used to measure various phenotypic parameters like shoot biomass, yield and leaf morphology, and salinity tolerance [26, 27]. Hyperspectral imaging is another widely used in plant phenotyping. In this technique, sensors capture multiple spectral ranges, enabling detailed measurements for various phenotypic parameters. Notable applications of hyperspectral imaging include estimating nutrient content and detecting plant responses to abiotic or biotic stressors [28]. The other imaging techniques are not as widely used as visible or hyperspectral imaging; however, they are also effective in measuring plant phenotypic parameters. For instance, thermal imaging in plant water stress relations [9, 16], fluorescence imaging in used plant pathogenic symptom monitoring [29, 30], and tomography in estimating plant organ water content [31].

2.2 Processing Techniques

There is a lack of generalised or standardised image-processing techniques in plant phenotyping; hence, bespoke image-processing pipelines are designed for specific plant species or genotypes like Arabidopsis [9, 32]. Therefore, various phenotyping systems' image processing and analysis pipelines can vastly differ; however, as mentioned earlier in the review of plant phenotyping traits, various morphological traits are derived from specific organs or parts of the plant. Furthermore, segmentation methods directly affect the accuracy of more complex tasks like object detection, 3D reconstruction, and classification. Hence, the validity and preciseness of the derived phenotypical measurements and analysis depend on the image segmentation method selected [33, 34].

In plant phenotyping, appropriate selection of segmentation techniques is vital when considering the type of phenotypic parameter to be measured. Segmentation methods in plant phenotyping are typically classifiable as traditional or learning-based [33]. Traditional segmentation methods in plant phenotyping typically target holistic phenotypical traits like plant height and include frame differencing [35], colour-based [36], edge detection [37], spectral band difference-based [38], shape modelling-based [39], and graph-based segmentation [40]. In contrast to traditional techniques, learning-based methods are more suitable for targeting component-based traits of specific organs or parts of a plant [33]. Many learning-based methods use deep learning and machine-learning mechanisms to handle segmentation [41, 42].

3 Low-cost Automated Image-Based Plant Phenotyping In Greenhouses

There is a growing need for low-cost, flexible, modifiable, and extendable automated image processing systems for plant phenotyping applications. Many automated phenotypical image processing systems in greenhouses employ various advanced technologies, such as robotics [43], automated conveyor belts, and benchtop configurations [44, 45], and advanced image sensors in high-throughput phenotyping systems [46]. However, these solutions are not accessible to all plant growers because they tend to be costly [47, 48, 49], and many systems are patented by specialized companies, rendering the hardware and software inflexible and unmodifiable [50]. Furthermore, existing low-cost phenotyping systems in greenhouses have bespoke image analysis and processing pipelines, which prevent them from being generalized and used in broader contexts [22, 9] .Therefore, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing systems for plant phenotyping applications.

3.1 Open-source Software, Systems and Affordable Hardware As Cost Reducers In Image-Based Plant Phenotyping

Recent studies in low-cost automated image-based plant phenotyping show increased development of open-source phenotyping systems and the use of readily accessible commercial off-the-shelf hardware [51, 49] to reduce the costs of image-based phenotyping in greenhouse environments. Cheap low-cost compute sensors called Raspberry PIs have been used with affordable RGB and hyperspectral sensors to monitor phenotypic traits in greenhouse environments [48]. Additionally, there has been an increase in open-source image-processing plant phenotyping systems like Phenobox [50].

3.2 Challenges In Automated Image-Based Plant Phenotyping

Effective choices in image sensors are crucial for developing low-cost, automated image-based phenotyping systems [12]. Modern image sensors in image-based phenotyping generate massive amounts of data [52], necessitating the development of efficient image processing pipelines in these contexts to handle large datasets in the shortest possible time while also delivering precise and accurate phenotypical analysis [49].

The choice of sensor has a non-trivial influence on the image analysis and processing techniques used in plant phenotyping and additionally also determines the breadth of traits that can be analysed [9]. For example, hyperspectral imaging is practical for studying plant metabolic status; however, the sensors required generate Terabytes of data when monitoring multiple plants [53]. Consequently, in low-cost settings, hyperspectral imaging of component-based phenotypical traits becomes infeasible because learning-based image techniques, which are most suitable for this purpose, are costly, time-consuming, and impractical when large amounts of phenotypical data are processed [53]. Additionally, low-cost hardware has constrained bandwidth, storage, and compute power. Therefore, remote cloud image processing and analysis are required [54], which can be expensive [49].

In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity traditional segmentation methods [49], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [48], and image compression techniques [55]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities. Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [33]. RGB sensors are affordable and generate less raw data than other methods, such as hyperspectral imaging; however, their use is limited to the phenotyping of physiological plant traits [9]. Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [55]. Therefore, a delicate balance must be struck between cost and numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [34].

3.3 Potential Advances For Low-Cost Automated Image-Based Plant Phenotyping

Low-cost plant phenotyping is not a nascent field of inquiry. However, many tracts of innovation have not been fully exploited. For instance, learning-based imaging techniques for analysis and processing can reduce analysis and segmentation costs but struggle when executed on resource-constrained hardware [24]. Therefore, there is a need for the development of efficient and optimised deep-learning or machine-learning models in resource-constrained environments. Similarly, many existing phenotyping systems are not generalizable to other plant species [9, 32] and rely on proprietary software or hardware [50]. Hence, there are opportunities to develop modifiable and flexible image-processing pipelines that are amenable to a broader variety of plant phenotyping contexts. Additionally, there are not many mobile applications for plant phenotyping. Mobile devices have advanced image sensors and can locally run image analysis and processing tasks, eliminating the need for cloud computing resources that increase phenotyping costs [34]. In short, numerous opportunities exist to improve the overall affordability and generalisability of plant phenotyping in low-cost settings.

4 Conclusion

Ensuring an adequate food supply in response to the growing human population is crucial [2], and image-based plant phenotyping has proven to be an effective tool in developing cultivars with maximum yield and resistance to various environmental stressors [4]. These methods are non-destructive, more time-efficient, replicable, and affordable than traditional methods [11]. However, image sensors in plant phenotyping generate large amounts of data, which necessitate robust image analysis and processing tools to handle this data effectively [52]. Large plant data throughput makes image-based plant phenotyping in automated greenhouse environments cost-prohibitive, as affordable off-the-shelf hardware and sensors often lack adequate local resources to store or process large amounts of data [53]. Consequently, remote cloud computing solutions handle image processing and analysis tasks [54], which can further increase operational costs in greenhouse management [49]. Image-data compression techniques, low-complexity image analysis and processing techniques, and low-cost off-the-shelf hardware are effective in reducing the cost of image-based plant phenotyping by reducing the amount of computation and data throughput required [55, 48, 49]. However, these solutions come with marked trade-offs in terms of accuracy, precision, depth, and breadth of plant trait analysis. Moreover, existing solutions are not generalizable to other environments and plant species [22, 9]. In short, there is a need for low-cost, open-source, flexible, modifiable, and robust plane phenotyping image processing and analysis tools that are effective and efficient on resource-constrained hardware [49]. Future research should develop low-cost image processing tools that are precise, accurate, and adaptable to various plant monitoring contexts or parameters. In doing so, automated image-based techniques for plant monitoring will become more accessible in low-cost greenhouse environments.

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